

Theophylline population PK model

Methods

Theophylline concentrations from 12 subjects were described by a one-compartment model with first-order absorption (Figure 1), estimated with SAEM in nlmixr2.

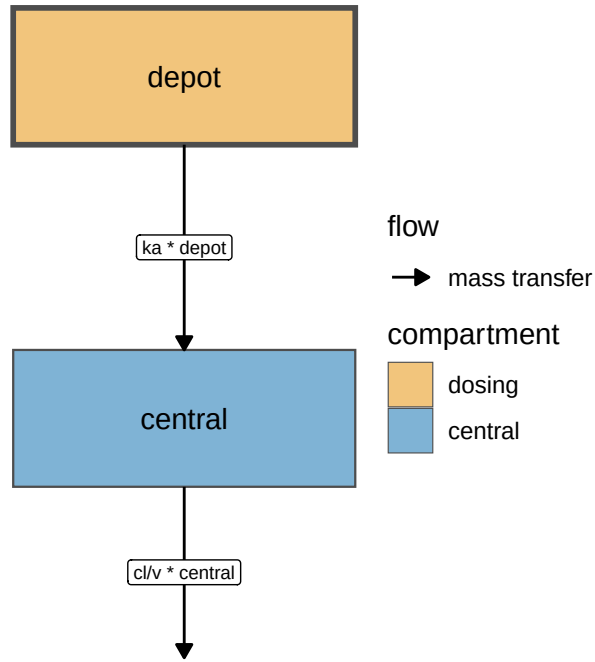


Figure 1: Structure of the final model.

The model is defined by the following equations:

$$\begin{aligned}ka &= \exp(tka + eta.ka) \\cl &= \exp(tcl + eta.cl) \\v &= \exp(tv + eta.v) \\ \frac{d \text{depot}}{dt} &= -ka \times \text{depot} \\ \frac{d \text{central}}{dt} &= ka \times \text{depot} - \frac{cl}{v} \times \text{central} \\cp &= \frac{\text{central}}{v} \\cp &\sim \text{add}(\text{add.sd})\end{aligned}$$

Results

The objective function value was 123.05. Table 1 lists the final parameter estimates.

Table 1: Final parameter estimates.

	Back-transformed	CI Lower	CI Upper	%RSE	BSV(CV%)
tka	1.586	1.089	2.308	41.554	71.516
tcl	2.750	2.356	3.210	7.803	26.159
tv	31.604	28.920	34.538	1.311	14.351
add.sd	0.697	0.604	0.789	6.772	